UNIT

THREE

Geometry and measurement in three dimensions

Exercise 8 The 3-dimensional orthogonal coordinate system.

Exercise 9 Vectors in the space.

Exercise 10 Scalar product of two vectors.

Exercise 11 Vector product and scalar triple product.



Exercise 8

On the 3-dimensional orthogonal coordinate system

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Determine the position of each of the following points using 3-dimensional orthogonal coordinate system :**

(1) A (3 , 4 , 5) (2) B (– 3 , 2 , 5)

2 **Find AB in each of the following :**

(1) A (7 , 0 , 4) , B (1 , 0 , 0)

(2) A (4 , 1 , 9) , B (2 , 1 , 6)

(3) A (1 , 1 , – 7) , B (– 2 , – 3 , – 7) « $2\sqrt{13}$, $\sqrt{13}$, 5 »

3 **Find the coordinates of the midpoint of $\overline{AB}$ where :**

(1) A (1 , – 3 , 2) , B (4 , – 1 , 4) (2) A (– 3 , 5 , 5) , B (– 6 , 4 , 8)

4 **Choose the correct answer from those given :**

(1) The point A (2 , – 3 , 0) lies .....

(a) on the z-axis. (b) in the y z-plane. (c) in the $\mathcal{X}$ y-plane. (d) on the $\mathcal{X}$-axis.

(2) The point A (2 , 0 , 0) lies .....

(a) on the y-axis. (b) on the z-axis. (c) on the $\mathcal{X}$-axis. (d) in the y z-plane.

(3) The distance between the point (2 , – 3 , 5) and the $\mathcal{X}$ z-plane equals ..... length unit.

(a) 2 (b) – 3 (c) 3 (d) 5

(4) The distance between the point (a , b , c) and y-axis equals .....

(a) $\sqrt{a^2 + c^2}$ (b) $\sqrt{a^2 + b^2}$ (c) $\sqrt{b^2 + c^2}$ (d) $\sqrt{a^2 + b^2 + c^2}$

(5) The perpendicular distance from the point (– 5 , – 3 , 4) to the $\mathcal{X}$-axis = ... length unit.

(a) 3 (b) 5 (c) 4 (d) 10

(6) If the point ($\mathcal{X}$, y , z) lies in the $\mathcal{X}$ z-plane , then .....

(a) $\mathcal{X} = 0$ (b) y = 0 (c) z = 0 (d) $\mathcal{X} + y = 0$

(7) x y-plane and y z-planes intersect at .....

(a) origin point. (b) x-axis. (c) y-axis. (d) z-axis.

(8) The coordinate planes x y , x z , z y intersecting at .....

(a) the origin. (b) x-axis. (c) y-axis. (d) z-axis.

(9) The straight lines $\overleftrightarrow{xx}$, $\overleftrightarrow{zz}$ form the coordinate plane whose equation is .....

(a) $x = 0$ (b) $y = 0$ (c) $z = 0$ (d) $y = 2$

(10) The equation of z-axis in the space is .....

(a) $x = 0 , y = 0$ (b) $x = 0 , z = 0$ (c) $y = 0 , z = 0$ (d) $x = 0$

(11) The midpoint of the line segment $\overline{DE}$ where D $(2 , 3 , 3)$, E $(6 , -1 , -4)$ is .....

(a) $(4 , 2 , 3\frac{1}{2})$ (b) $(2 , 1 , \frac{1}{2})$ (c) $(4 , 1 , -\frac{1}{2})$ (d) $(4 , 1 , \frac{1}{2})$

(12) If A (a , b , c) is the midpoint of $\overline{DE}$ where D $(-4 , 0 , 5)$ and E $(-2 , 4 , -13)$ then $a + b + c =$ .....

(a) -5 (b) -6 (c) 3 (d) 4

(13) *(2nd Session 2024)* If A $(2m - 1 , 3m , 2m + 1)$, B $(3 - 2m , 4 - 3m , 5 - 2m)$, then the midpoint of $\overline{AB}$ is .....

(a) $(1 , 3 , 6)$ (b) $(1 , 2 , 3)$ (c) $(5 , 4 , 3)$ (d) $(1 , 4 , 6)$

(14) If the midpoint of $\overline{AB}$ lies in the cartesian plane x z and A $(-3 , 12 + k , 5)$, B $(1 , 3k , -2)$, then $k =$ .....

(a) 5 (b) -3 (c) -2 (d) 1

(15) If A $(7 , -1 , 8)$, B $(11 , 2 , -4)$ then the length of $\overline{AB} =$ ..... cm.

(a) 10 (b) 11 (c) 12 (d) 13

(16) The point $(2 , 0 , -3)$ lies in the coordinates plane whose equation is .....

(a) $z = 0$ (b) $y = 0$ (c) $x = 0$ (d) $x + y = -1$

(17) If the point $(2a , a + 3 , 5)$ lies in the cartesian plane x z , then its distance from y z-plane equals ..... unit length.

(a) 3 (b) 5 (c) 6 (d) zero

(18) If the point $(a - 2 , 5 , a - 4)$ at a distance 5 units from the y z-plane and at a distance 3 units from the x y-plane , then $a =$ .....

(a) 2 (b) 4 (c) 7 (d) 7 or -3

(19) If $(5 , 6 , -3)$ is the midpoint of $\overline{AB}$ where A $(3 , -1 , 5)$, then B = ...

(a) $(4 , \frac{5}{2} , 1)$ (b) $(7 , 13 , -11)$ (c) $(-2 , -7 , 8)$ (d) $(3 , 2 , 13)$

(20) The distance between the point $(-3, 4, 5)$ from the origin = ... length unit.

(a) 5 (b) $5\sqrt{2}$ (c) 10 (d) 25

(21) If A $(-4, -2, 3)$, B $(1, 2, k)$ and the length of $\overline{AB} = \sqrt{77}$, then k = ...

(a) -3 or 6 (b) -3 or 12 (c) 9 or 6 (d) 9 or -3

(22) All points in space in the form $(x, 5, z)$ lies in the plane whose equation .....

(a) $x = 5$ (b) $y = 5$ (c) $z = 0$ (d) $y = 0$

(23) If the points A $(6, 0, 3)$, B $(7, 1, 7)$, C $(9, 3, 15)$ lie on the same straight line , then A divides $\overline{BC}$ in the ratio .....

(a) 3 : 1 internally. (b) 1 : 3 externally. (c) 2 : 3 internally. (d) 1 : 2 externally.

(24) ***(Speciman 2025)*** If the perpendicular distance between the point $(-3, 4, a)$ and the y-axis equals 5 units , then a = ...

(a) 5 (b) ± 4 (c) ± 3 (d) 12

(25) If the point $(k, 3, -5)$ equidistant from x-axis and y-axis , then k = ...

(a) ± 2 (b) ± 3 (c) ± 4 (d) ± 5

(26) If the midpoint of $\overline{AB} \in x$-axis where A $(2, 12 + k, k)$, B $(4, m, 8 - m)$, then $k - 3m$ = ...

(a) 4 (b) -4 (c) -2 (d) -10

(27) The point A $(3, -5, 1)$ in the space , then the sum of its distances from the three coordinate planes = ... length units.

(a) 2 (b) 1 (c) 9 (d) 35

(28) The image of the point $(-2, 3, 4)$ by reflection in the z-axis is .....

(a) $(-2, 3, 4)$ (b) $(2, 3, 4)$ (c) $(2, -3, -4)$ (d) $(2, -3, 4)$

(29) If M $(1, 2, 3)$, N $(4, 2, 3)$, R $(1, 6, 3)$ are the midpoints of $\overline{AB}$, $\overline{BC}$, $\overline{CA}$ respectively , then the perimeter of Δ ABC is ..... length unit.

(a) 12 (b) 13 (c) 14 (d) 24

(30) ***(1st Session 2022)*** If A $(2, -5, 7)$, B $(1, 3, 6)$ and C is a point belongs to y-axis and equidistant from A and B , then the point C is .....

(a) $(0, 2, 0)$ (b) $(0, 1, 0)$ (c) $(0, -2, 0)$ (d) $(0, -1, 0)$

(31) ***(2nd Session 2022)*** If A , B and C are three points in space , where A $(3, 4, 8)$, B $(1, 2, 7)$ and C $(3, 0, 6)$, then .....

(a) 2 AB = BC (b) AB = 2 BC (c) AB = BC (d) 3 AB = BC

(32) If L , M , N are the projections of the point A $(3, 4, 5)$ on the x-axis , y-axis and z-axis respectively , then M = ...

(a) $(3, 0, 5)$ (b) $(0, 4, 0)$ (c) $(0, 4, 5)$ (d) $(0, -4, 0)$

(33) If L , M , N are the projections of the point A (3 , 4 , 5) on the $\mathcal{X}$ y-plane , y z-plane and $\mathcal{X}$ z-plane respectively , then L = ...

(a) (– 3 , – 4 , 0) (b) (0 , 4 , 5) (c) (3 , 4 , 0) (d) (3 , 0 , 0)

(34) If the point E is equidistant from the points O (0 , 0 , 0) , A (ℓ , 0 , 0) , B (0 , m , 0) and C (0 , 0 , n) , then the point E = ...

(a) (ℓ , m , n) (b) $\left(\frac{-\ell}{2} , \frac{-m}{2} , \frac{-n}{2}\right)$ (c) ($-\ell$, – m , – n) (d) $\left(\frac{\ell}{2} , \frac{m}{2} , \frac{n}{2}\right)$

(35) The point lies on the y-axis and at a distance $\sqrt{10}$ length units from the point (1 , 2 , 3) is .....

(a) (0 , – 2 , 0) (b) (0 , 2 , 0) (c) (0 , 4 , 0) (d) (0 , – 4 , 0)

(36) If A (6 , – 2 , 4) , B (2 , 4 , – 8) , C (– 2 , 2 , 4) are three consecutive vertices of a parallelogram ABCD , then D = ...

(a) (2 , $\frac{4}{3}$, 0) (b) (– 2 , 4 , – 16) (c) (2 , – 4 , 16) (d) (1 , – 2 , 8)

(37) *(2nd Session 2021)* If ABC is a triangle in which D is the midpoint of $\overline{BC}$, A (3 , 1 , 5) , B (2 , 3 , 7) , C (0 , 3 , 1) , then the length of $\overline{AD}$ = ... length unit.

(a) 9 (b) 2 (c) 7 (d) 3

(38) *(Speciman 2025)* If the triangle with vertices (4 , 5 , 2) , (1 , k , 3) , (2 , 4 , 5) is an equilateral triangle , then k ∈ .....

(a) $\left\{\frac{21}{5}\right\}$ (b) {7} (c) {3 , 7} (d) $\left\{7 , -\frac{21}{5}\right\}$

(39) *(Trial 2021)* If A (3 , – 4 , 0) , B (15 , 0 , 2) , C (0 , – 8 , 4) are three points in the space and they form triangle ABC , then the distance between its centroid and the $\mathcal{X}$ z-plane is .....

(a) greater than the distance from the $\mathcal{X}$ y-plane.

(b) smaller than or equal to the distance from the $\mathcal{X}$ y-plane.

(c) greater than the distance from the y z-plane.

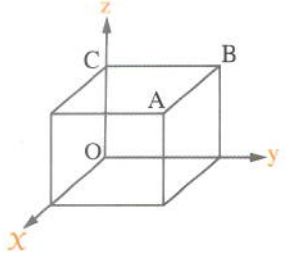
(d) greater than or equal to the distance from the y z-plane.

(40) The opposite figure is a cuboid : A (5 , 8 , 4) , then

First : The point B is .....

(a) (4 , 8 , 0) (b) (0 , 8 , 4)

(c) (5 , 8 , 0) (d) (5 , 0 , 4)



Second : The point C is .....

(a) (0 , 0 , 0) (b) (0 , 0 , 4)

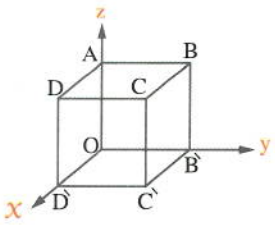
(c) (5 , 0 , 4) (d) (0 , 8 , 0)

(41) ABCDOB̀C̀D̀ is a cube of edge length 5 length unit , then

First : The point C is .....

(a) (5 , 5 , 0) (b) (5 , 5 , 5)

(c) (0 , 0 , 5) (d) (0 , 5 , 0)



Second : The point D̀ is .....

(a) (0 , 0 , 5) (b) (5 , 5 , 0)

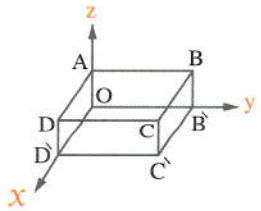
(c) (0 , 5 , 5) (d) (5 , 0 , 0)

Third : The diagonal length of the cube = ... length units.

(a) $5\sqrt{2}$ (b) $5\sqrt{3}$ (c) 5 (d) $5\sqrt{6}$

(42) In the opposite figure :

A cuboid in which C̀ (5 , 8 , 0) , D (5 , 0 , 3) , then



First : The point C is .....

(a) (5 , 8 , 0) (b) (5 , 3 , 8)

(c) (5 , 8 , 3) (d) (5 , 8 , 8)

Second : The volume of the cuboid is ..... cubic units.

(a) 64 (b) 120 (c) 144 (d) 150

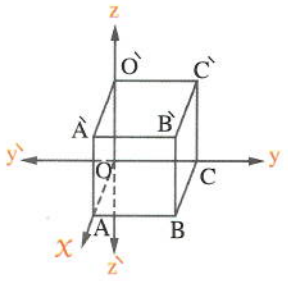
Third : The equation of the plane OB̀C̀D̀ is .....

(a) $x = 0$ (b) $y = 0$ (c) $z = 0$ (d) $z = 3$

Fourth : The equation of the plane DD̀C̀C is .....

(a) $x = 0$ (b) $y = 0$ (c) $x = 5$ (d) $z = 3$

5 The opposite figure represents a cube whose volume is 27 cubic unit and one of its vertices is the origin find coordinates of the other vertices.



6 Find the coordinates of the point A in each of the following cases :

(1) A (1 – k , 2 k , 3 + k) lies in x y-plane.

(2) A (0 , m – 3 , m + 4) lies on z-axis.

(3) A (n – 1 , n + 4 , 5 n) lies in the plane $y = 5$

(4) A (ℓ + 5 , 2 ℓ , ℓ) at a distance $2\sqrt{5}$ length unit from x-axis.

(5) A (m , 2 m – n , m + n + 3) $\in x$-axis.

7 Prove that the points A (7 , 1 , 4) , B (8 , 2 , 8) , C (10 , 4 , 16) are collinear.

8 Prove that the points A (1 , 2 , 5) , B (2 , 5 , 3) , C (– 1 , 3 , 2) are vertices of an equilateral triangle and find its area. « $\frac{7\sqrt{3}}{2}$ square unit »

9 Prove that the points A (2 , – 1 , 3) , B (– 4 , 4 , 2) , C (– 2 , 5 , 1) are vertices of right-angled triangle and find its area. « $2\sqrt{21}$ square units »

10 Prove that the points A (7 , 1 , 3) , B (5 , 3 , 4) , C (3 , 5 , 3) are vertices of an isosceles triangle.

11 If the points A (5 , 1 , 3) , B (7 , 3 , k) , C (5 , 5 , 3) are the vertices of a triangle , prove that this triangle is isosceles , then find the real values of k which makes the triangle equilateral. « $3 \pm 2\sqrt{2}$ »

12 If C (– 1 , 4 , 0) is the midpoint of $\overline{AB}$ where B (4 , – 2 , 1)
Find coordinates of point A « (– 6 , 10 , – 1) »

13 If C (– 1 , 6 , – 5) is the midpoint of $\overline{AB}$ where A (k – 2 , – 1 , m + 3) , B (2 , n – 7 , – 2) find the value of k + m – n « – 33 »

14 Find the perimeter of the triangle formed from joining the midpoints of the sides of Δ ABC where A (2 , 5 , – 1) , B (6 , – 3 , 3) , C (– 4 , 7 , 1) « 15.4 »

15 If A ∈ X-axis , B ∈ y-axis , C ∈ z-axis and if point (1 , – 1 , 0) is the midpoint of $\overline{AB}$ and point (0 , – 1 , 2) is the midpoint of $\overline{BC}$, find the midpoint of $\overline{AC}$ « (1 , 0 , 2) »

16 If A , B , C , D are 4 coplanar points , **prove that :**
(1) A (0 , 4 , 1) , B (2 , 3 , – 1) , C (4 , 5 , 0) , D (2 , 6 , 2) are vertices of a square.
(2) A (5 , – 1 , 1) , B (7 , – 4 , 7) , C (1 , – 6 , 10) , D (– 1 , – 3 , 4) are vertices of a rhombus.

17 Let the points E , F , D be the midpoints of the sides of Δ ABC , $\overline{AB}$, $\overline{BC}$, $\overline{AC}$ respectively , E (1 , 2 , 6) , F (0 , 3 , 2) , D (2 , 0 , – 2).
Find the coordinates of the vertices of Δ ABC